FACULTE DES SCIENCES ET TECHNOLOGIES

Sciences Informatiques

Cours : Programmation Python

TD Nº 4: Python

Blomy ANTOINE

L3

Sous la direction du professeur :
Ismaël SAINT-AMOUR

18 MARS 2026

Exo1 : MATRICES

```
20 mars 15:13
blomy-antoine@blomy-antoine-NE56R: ~/TP4 — nano exo1.py
~/TP4
  GNU nano 8.4                                exo1.py
import numpy as np

A = np.array([
    [1,2,3,4],
    [5,6,7,8],
    [9,10,11,12],
    [13,14,15,16]
])

B = np.array([
    [2,0,1,3],
    [1,2,0,4],
    [3,1,2,1],
    [0,2,3,2]
])

print("Addition :\n", A + B)
print("\nProduit élément par élément :\n", A * B)
print("\nProduit matriciel :\n", A @ B)

^G Aide      ^O Écrire      ^F Chercher    ^K Couper      ^T Exécuter    ^C Emplacement  M-U Annuler     M-A Marquer     M-] -> Crochet  M-B Précédent
^X Quitter   ^R Lire fich.  ^\ Remplacer   ^U Coller      ^J Justifier   ^/ Aller ligne   M-E Refaire    M-6 Copier      ^B Retrouver    M-F Suivant
```

```
blomy-antoine@blomy-antoine-NE56R: ~/TP4
~/TP4
Using cached pillow-12.1.1-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (7.0 MB)
Using cached pyparsing-3.3.2-py3-none-any.whl (122 kB)
Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Downloading scipy-1.17.1-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (35.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 35.2/35.2 MB 474.2 kB/s eta 0:00:00
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Downloading threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Installing collected packages: threadpoolctl, six, pyparsing, pillow, packaging, numpy, kiwisolver, joblib, fonttools, cycler, scipy, python-dateutil, contourpy
, scikit-learn, matplotlib
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.62.1 joblib-1.5.3 kiwisolver-1.5.0 matplotlib-3.10.8 numpy-2.4.3 packaging-26.0 pillow-12.1.1 p
yparsing-3.3.2 python-dateutil-2.9.0.post0 scikit-learn-1.8.0 scipy-1.17.1 six-1.17.0 threadpoolctl-3.6.0
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ python exo5.py
/home/blomy-antoine/TP4/exo5.py:16: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be shown
  plt.show()
Pente : -0.5
Intercept : 105.0
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ nano exo1,py
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ nano exo1.py
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ python exo1.py
Addition :
 [[ 3  2  4  7]
 [ 6  8  7 12]
 [12 11 13 13]
 [13 16 18 18]]

Produit élément par élément :
 [[ 2  0  3 12]
 [ 5 12  0 32]
 [27 10 22 12]
 [ 0 28 45 32]]

Produit matriciel :
 [[ 13  15  19  22]
 [ 37  35  43  62]
 [ 61  55  67 102]
 [ 85  75  91 142]]
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$
```

exo 2 : **PROBABILITÉ**

```
blomy-antoine@blomy-antoine-NE56R: ~/TP4 — nano exo2.py
~/TP4

  GNU nano 8.4                                exo2.py
filles = 35
garcons = 25

reussite_filles = 18
reussite_garcons = 12

total = filles + garcons
reussite = reussite_filles + reussite_garcons

proba = reussite / total

print("Probabilité de réussite :", proba)

                                [ Lecture de 12 lignes ]
^G Aide      ^O Écrire    ^F Chercher  ^K Couper    ^T Exécuter  ^C Emplacement  M-U Annuler   M-A Marquer   M-] -> Crochet  M-B Précédent
^X Quitter   ^R Lire fich. ^\ Remplacer ^U Coller   ^J Justifier ^/ Aller ligne   M-E Refaire  M-6 Copier    ^B Retrouver    M-F Suivant
```

```
blomy-antoine@blomy-antoine-NE56R: ~/TP4
~/TP4

Downloading scipy-1.17.1-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (35.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 35.2/35.2 MB 474.2 kB/s eta 0:00:00
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Downloading threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Installing collected packages: threadpoolctl, six, pyparsing, pillow, packaging, numpy, kiwisolver, joblib, fonttools, cycler, scipy, python-dateutil, contourpy
, scikit-learn, matplotlib
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.62.1 joblib-1.5.3 kiwisolver-1.5.0 matplotlib-3.10.8 numpy-2.4.3 packaging-26.0 pillow-12.1.1 p
yparsing-3.3.2 python-dateutil-2.9.0.post0 scikit-learn-1.8.0 scipy-1.17.1 six-1.17.0 threadpoolctl-3.6.0
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ python exo5.py
/home/blomy-antoine/TP4/exo5.py:16: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be shown
  plt.show()
Pente : -0.5
Intercept : 105.0
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ nano exo1,py
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ nano exo1.py
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ python exo1.py
Addition :
 [[ 3  2  4  7]
 [ 6  8  7 12]
 [12 11 13 13]
 [13 16 18 18]]

Produit élément par élément :
 [[ 2  0  3 12]
 [ 5 12  0 32]
 [27 10 22 12]
 [ 0 28 45 32]]

Produit matriciel :
 [[ 13  15  19  22]
 [ 37  35  43  62]
 [ 61  55  67 102]
 [ 85  75  91 142]]
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ nano exo2.py
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ python exo2.py
Probabilité de réussite : 0.5
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$
```

exo 3 : PROBABILITÉ CONDITIONNELLE

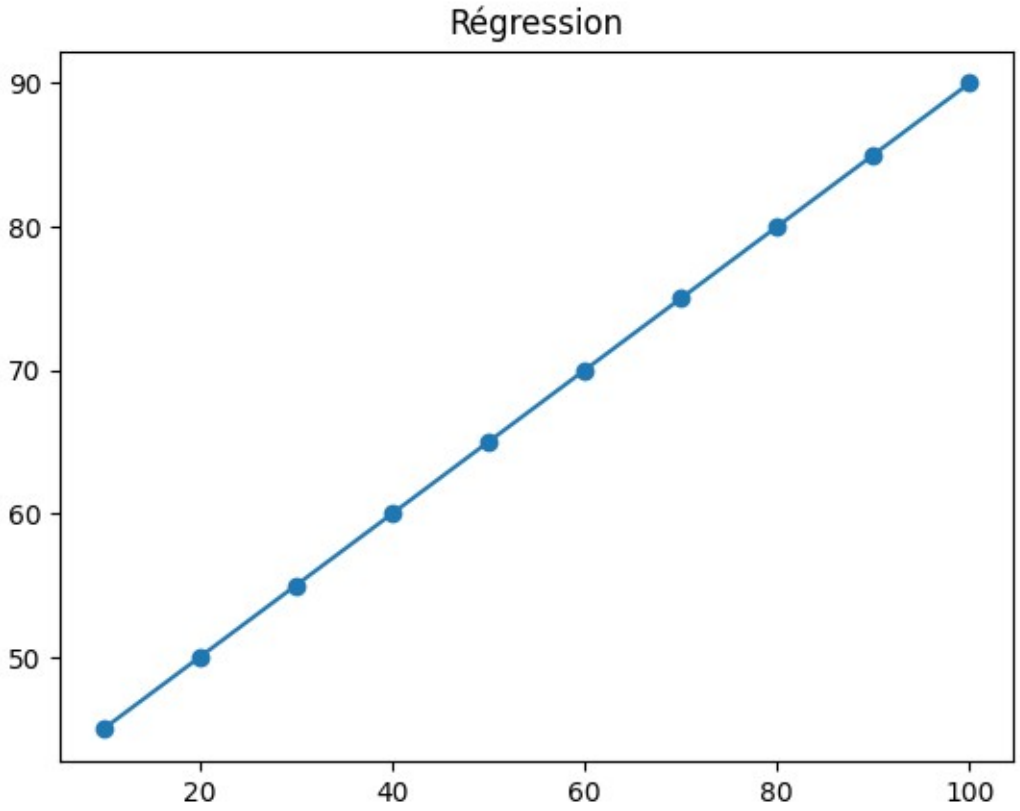
```
blomy-antoine@blomy-antoine-NE56R: ~/TP4 — nano exo3.py
~/TP4

  GNU nano 8.4                                   exo3.py
reussite_filles = 18
reussite_garcons = 12

total_reussite = reussite_filles + reussite_garcons

proba = reussite_filles / total_reussite

print("Probabilité (fille | réussite) :", proba)

                                   [ Lecture de 8 lignes ]
^G Aide      ^O Écrire    ^F Chercher  ^K Couper    ^T Exécuter  ^C Emplacement  M-U Annuler   M-A Marquer   M-] -> Crochet  M-B Précédent
^X Quitter   ^R Lire fich. ^\ Remplacer ^U Coller    ^J Justifier ^/ Aller ligne  M-E Refaire   M-6 Copier    ^B Retrouver    M-F Suivant
```

```
blomy-antoine@blomy-antoine-NE56R: ~/TP4
~/TP4

Downloading threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Installing collected packages: threadpoolctl, six, pyparsing, pillow, packaging, numpy, kiwisolver, joblib, fonttools, cycler, scipy, python-dateutil, contourpy
, scikit-learn, matplotlib
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.62.1 joblib-1.5.3 kiwisolver-1.5.0 matplotlib-3.10.8 numpy-2.4.3 packaging-26.0 pillow-12.1.1 p
yparsing-3.3.2 python-dateutil-2.9.0.post0 scikit-learn-1.8.0 scipy-1.17.1 six-1.17.0 threadpoolctl-3.6.0
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ python exo5.py
/home/blomy-antoine/TP4/exo5.py:16: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be shown
  plt.show()
Pente : -0.5
Intercept : 105.0
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ nano exo1,py
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ nano exo1.py
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ python exo1.py
Addition :
 [[ 3  2  4  7]
 [ 6  8  7 12]
 [12 11 13 13]
 [13 16 18 18]]

Produit élément par élément :
 [[ 2  0  3 12]
 [ 5 12  0 32]
 [27 10 22 12]
 [ 0 28 45 32]]

Produit matriciel :
 [[ 13  15  19  22]
 [ 37  35  43  62]
 [ 61  55  67 102]
 [ 85  75  91 142]]
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ nano exo2.py
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ python exo2.py
Probabilité de réussite : 0.5
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ nano exo3.py
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$ python exo3.py
Probabilité (fille | réussite) : 0.6
(env) blomy-antoine@blomy-antoine-NE56R:~/TP4$
```

exo 4 : EXERCICE 4 — RÉGRESSION (ÉDUCATION)

```
blomy-antoine@blomy-antoine-NE56R: ~/TP4 — nano exo4.py
~/TP4

  GNU nano 8.4                                exo4.py
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

X = np.array([10,20,30,40,50,60,70,80,90,100]).reshape(-1,1)
y = np.array([45,50,55,60,65,70,75,80,85,90])

model = LinearRegression()
model.fit(X, y)

y_pred = model.predict(X)

plt.scatter(X, y)
plt.plot(X, y_pred)
plt.title("Régression")
plt.show()

print("Pente :", model.coef_[0])
print("Intercept :", model.intercept_)

                                  [ Lecture de 19 lignes ]
^G Aide      ^O Écrire     ^F Chercher   ^K Couper     ^T Exécuter   ^C Emplacement  M-U Annuler   M-A Marquer   M-] -> Crochet  M-B Précédent
^X Quitter   ^R Lire fich. ^\ Remplacer  ^U Coller     ^J Justifier  ^/ Aller ligne   M-E Refaire  M-6 Copier    ^B Retrouver    M-F Suivant
```

```
blomy-antoine@blomy-antoine-NE56R: ~
~
Collecting pyparsing>=3 (from matplotlib)
  Using cached pyparsing-3.3.2-py3-none-any.whl.metadata (5.8 kB)
Collecting python-dateutil>=2.7 (from matplotlib)
  Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting scipy>=1.10.0 (from scikit-learn)
  Using cached scipy-1.17.1-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (62 kB)
Collecting joblib>=1.3.0 (from scikit-learn)
  Using cached joblib-1.5.3-py3-none-any.whl.metadata (5.5 kB)
Collecting threadpoolctl>=3.2.0 (from scikit-learn)
  Using cached threadpoolctl-3.6.0-py3-none-any.whl.metadata (13 kB)
Collecting six>=1.5 (from python-dateutil>=2.7->matplotlib)
  Using cached six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Using cached numpy-2.4.3-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (16.6 MB)
Using cached matplotlib-3.10.8-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (8.7 MB)
Using cached scikit_learn-1.8.0-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (8.9 MB)
Using cached contourpy-1.3.3-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (362 kB)
Using cached cycler-0.12.1-py3-none-any.whl (8.3 kB)
Using cached fonttools-4.62.1-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (5.0 MB)
Using cached joblib-1.5.3-py3-none-any.whl (309 kB)
Using cached kiwisolver-1.5.0-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (1.5 MB)
Using cached packaging-26.0-py3-none-any.whl (74 kB)
Using cached pillow-12.1.1-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (7.0 MB)
Using cached pyparsing-3.3.2-py3-none-any.whl (122 kB)
Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Using cached scipy-1.17.1-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (35.2 MB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Using cached threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Installing collected packages: threadpoolctl, six, pyparsing, pillow, packaging, numpy, kiwisolver, joblib, fonttools, cycler, scipy, python-dateutil, contourpy
, scikit-learn, matplotlib
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.62.1 joblib-1.5.3 kiwisolver-1.5.0 matplotlib-3.10.8 numpy-2.4.3 packaging-26.0 pillow-12.1.1 p
yparsing-3.3.2 python-dateutil-2.9.0.post0 scikit-learn-1.8.0 scipy-1.17.1 six-1.17.0 threadpoolctl-3.6.0
(env) blomy-antoine@blomy-antoine-NE56R:~$ python exo4.py
/home/blomy-antoine/exo4.py:16: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be shown
  plt.show()
Pente : 0.5000000000000001
Intercept : 39.99999999999999
(env) blomy-antoine@blomy-antoine-NE56R:~$
```


Régression
90
80
70
60
50
20
40
60
80
100

exo 5 : PAUVRETÉ & EAU

```
blomy-antoine@blomy-antoine-NE56R: ~ — nano exo5.py

  GNU nano 8.4                                   exo5.py *
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

X = np.array([20,30,40,50,60,70,80,90,100]).reshape(-1,1)
y = np.array([95,90,85,80,75,70,65,60,55])

model = LinearRegression()
model.fit(X, y)

y_pred = model.predict(X)

plt.scatter(X, y)
plt.plot(X, y_pred)
plt.title("Pauvreté vs Eau")
plt.savefig("graphique.png")
plt.show()

print("Pente :", model.coef_[0])
print("Intercept :", model.intercept_)

^G Aide      ^O Écrire     ^F Chercher   ^K Couper     ^T Exécuter   ^C Emplacement  M-U Annuler    M-A Marquer    M-] -> Crochet  M-B Précédent
^X Quitter   ^R Lire fich. ^\ Remplacer  ^U Coller     ^J Justifier  ^/ Aller ligne   M-E Refaire    M-6 Copier     ^B Retrouver    M-F Suivant
```

```
blomy-antoine@blomy-antoine-NE56R: ~

Using cached numpy-2.4.3-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (16.6 MB)
Using cached matplotlib-3.10.8-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (8.7 MB)
Using cached scikit_learn-1.8.0-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (8.9 MB)
Using cached contourpy-1.3.3-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (362 kB)
Using cached cycler-0.12.1-py3-none-any.whl (8.3 kB)
Using cached fonttools-4.62.1-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (5.0 MB)
Using cached joblib-1.5.3-py3-none-any.whl (309 kB)
Using cached kiwisolver-1.5.0-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (1.5 MB)
Using cached packaging-26.0-py3-none-any.whl (74 kB)
Using cached pillow-12.1.1-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (7.0 MB)
Using cached pyparsing-3.3.2-py3-none-any.whl (122 kB)
Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Using cached scipy-1.17.1-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (35.2 MB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Using cached threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Installing collected packages: threadpoolctl, six, pyparsing, pillow, packaging, numpy, kiwisolver, joblib, fonttools, cycler, scipy, python-dateutil, contourpy
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Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.62.1 joblib-1.5.3 kiwisolver-1.5.0 matplotlib-3.10.8 numpy-2.4.3 packaging-26.0 pillow-12.1.1 p
yparsing-3.3.2 python-dateutil-2.9.0.post0 scikit-learn-1.8.0 scipy-1.17.1 six-1.17.0 threadpoolctl-3.6.0
(env) blomy-antoine@blomy-antoine-NE56R:~$ python exo4.py
/home/blomy-antoine/exo4.py:16: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be shown
  plt.show()
Pente : 0.5000000000000001
Intercept : 39.99999999999999
(env) blomy-antoine@blomy-antoine-NE56R:~$ nano exo4.py
(env) blomy-antoine@blomy-antoine-NE56R:~$ python exo4.py
/home/blomy-antoine/exo4.py:19: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be shown
  plt.show()
Pente : 0.5000000000000001
Intercept : 39.99999999999999
(env) blomy-antoine@blomy-antoine-NE56R:~$ nano exo5.py
(env) blomy-antoine@blomy-antoine-NE56R:~$ python exo5.py
/home/blomy-antoine/exo5.py:17: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be shown
  plt.show()
Pente : -0.5
Intercept : 105.0
(env) blomy-antoine@blomy-antoine-NE56R:~$
```

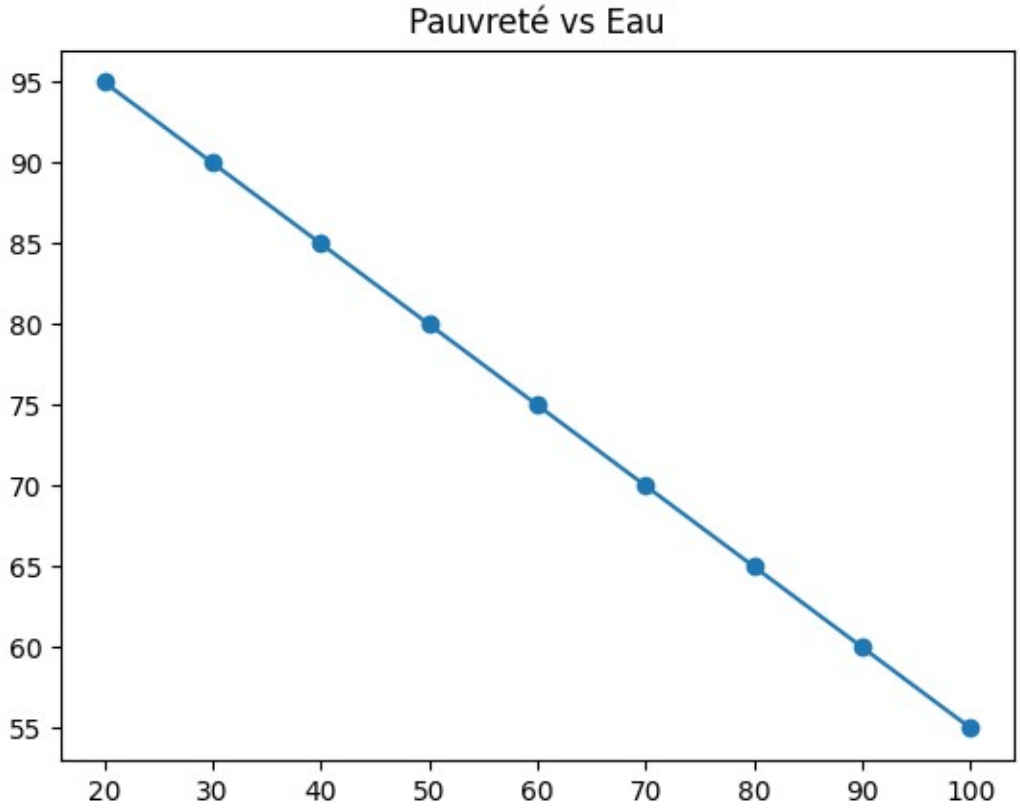
Pauvreté vs Eau
95
90
85
80
75
70
65
60
55
20
30
40
50
60
70
80
90
100